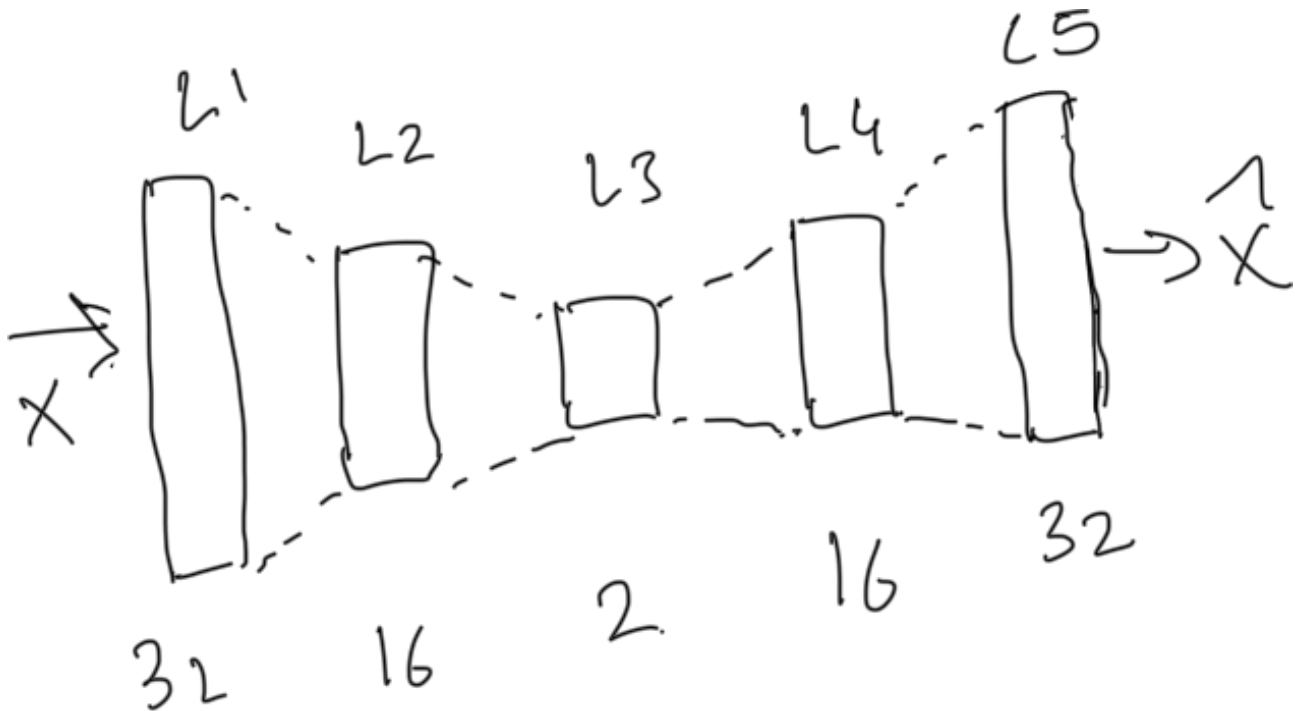


Consider a Neural Network (N) that takes as input a 20 Dimensional feature vector, X , and outputs a vector $\hat{X}$ such that $\hat{X}$ tries to be as close X . N is composed of 5 layers (L1 to L5) having the sizes as mentioned below.



Q. Suggest a suitable metric to make sure that $\hat{X}$ is as close X as possible.

Q. Code the above network in PyTorch. Each layer L1 to L5 is an `nn.linear` layer with ReLU non-linearity.

Note: Don't use the `nn.Sequential` construct to code the layers. At step of the training process (the forward function), return not just $\hat{X}$ but also the output of L3 (let's call it $\hat{z}$). $\hat{z}$ is called as the latent dimension for this Network.

There's no ReLU after L3 and L5.

Use `tensor.clone().detach()` to extract out z from L3.

Q. Train the network on the given data and observe its training error.

Q. Plot $\hat{z}$ in matplotlib and upload your plot.